

Research Methods - Prequel

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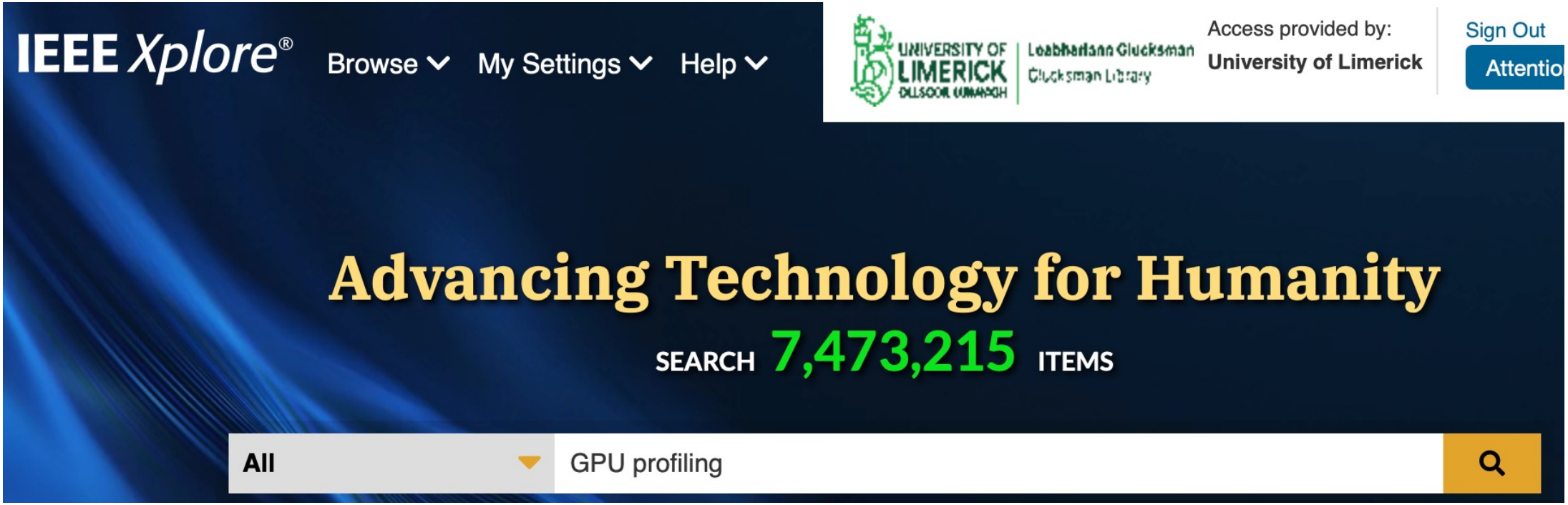
What are you doing for your FYP?

1. What is the problem? (Scope and challenge)
2. Who said it's a problem? (Contextual evidence)
3. What do we know about the problem? (Literature review)
4. How are you going to investigate the problem? (Method)
5. How is the state of knowledge changed? (= Why bother?)

How find relevant literature & make sense of it?

1. **Find**: Use a proper literature database
   → Today we use IEEExplore
   (there are others, e.g. ACM Digital Library, but IEEExplore is most targeted)
2. **Save**: Use a proper literature reference tool
   → Today we use Zotero (there are others but this is our fav)
3. **Make sense**: Write down what matters most in a note per paper
   1. What is the aim of the paper?
   2. How did the authors carry out the research, i.e. methods?
   3. What are the primary results?
   4. What do the authors think the results mean (for Q1)?
   5. What do you think it means?

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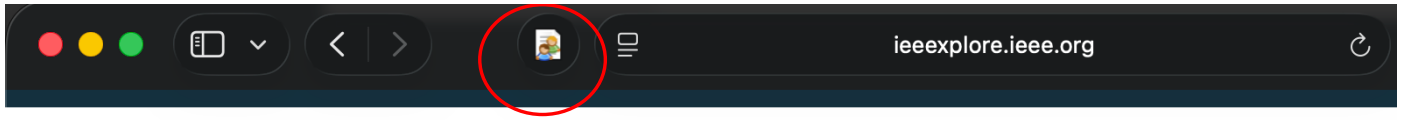
- [ ] Conferences (839)
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GTPin: Enhancing Intel GPU Profiling With High-Level Binary Instrumentation

Publisher: IEEE

Cite This | PDF

Konstantin Levit-Gurevich ; Alex Skaletsky ; Ortal Geller ; Moshe Goren ; Rawan Ziadat ; Rami Burstein

All Authors

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Full
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Abstract

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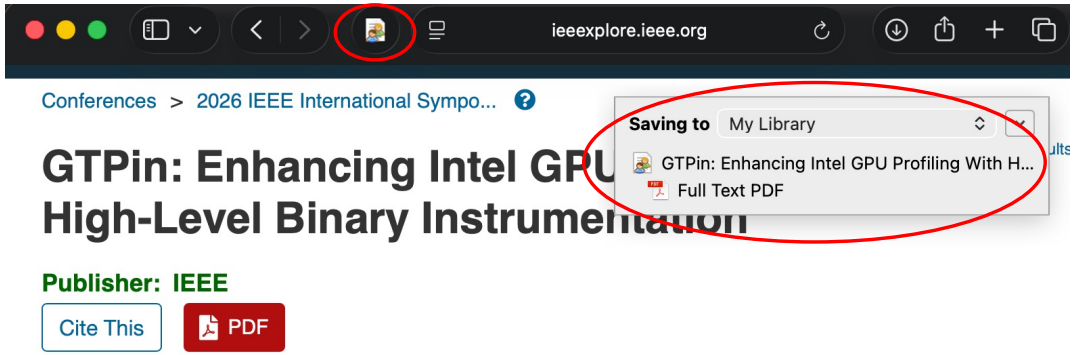
I. Introduction

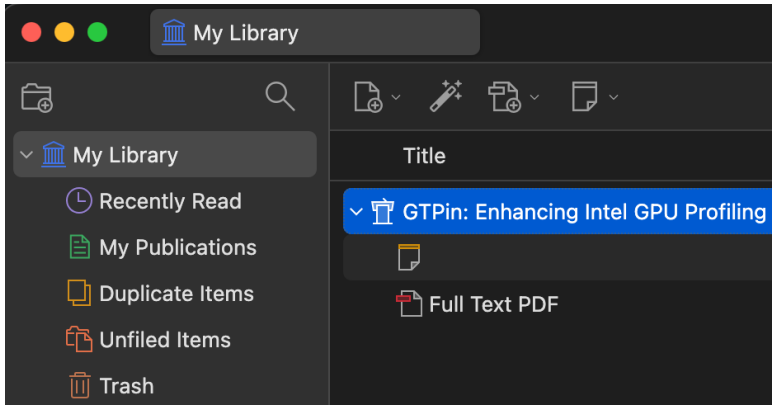
II. High-Level

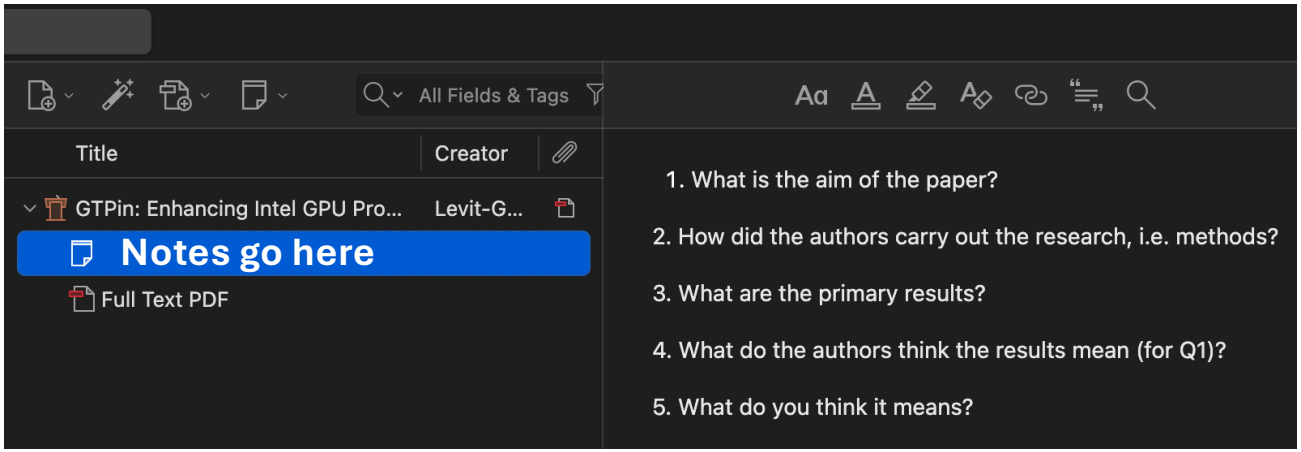
Abstract:
As GPUs are increasingly used for high-performance and data-intensive applications, optimizing workloads and memory behavior remains a significant challenge due to complex memory hierarchies, high parallelism, and limited runtime observability. Binary instrumentation has emerged as a key technique for analyzing and profiling GPU workloads

Zotero: Standalone & browser extension

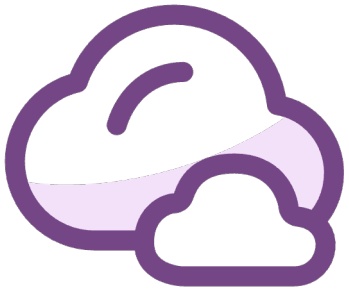








Do not edit
How to change the design



How many articles make a good lit review?

ⓘ The Slido app must be installed on every computer you're presenting from

slido



Targeted reading

How to (skim-)read 30 articles in 3 days



READ THE ABSTRACT & CONCLUSION



ANSWER LITERATURE NOTE QUESTIONS AS FAR AS YOU CAN



SKIM THE REST TO ANSWER REMAINING QUESTIONS

Best practices for better overview

- Is there a systematic literature review on the topic?
  (e.g. Jim Buckley on “Software Architecture Recovery”)
- Identify who the main authors of an area are
- Check if there is a “seminal” paper that many others reference
- Consider trying https://keenious.com for additional support

Repeater: What's your FYP?



Once you have skim-read that literature overview, revise your earlier proposal **introduction**:

1. "What is the problem?" (Scope and challenge)
2. "Who said it's a problem?" (Contextual evidence)
3. "What do we know about the problem?" (Literature review)
4. "How are you going to investigate the problem?" (Method)
5. "How is the state of knowledge changed?" (Impact)

→ Summarize that in a few sentences.

How to write the literature review

1. “What is the problem?” → Define the exact scope (in intro).

   Literature Review

2. Background: “Who said it’s a problem?”
   → Describe the contextual evidence to build foundation for what you want to do on top of what others have done.
3. Related Work: “What do we know about the problem?”
   → Describe how far others have come and what related results have been presented.
   → Differentiate the gap that remains (for you to answer).

Assignment



Install Zotero



Identify 30 relevant papers in IEEExplore & put them into library



Skim-read at least 3 and answer the note questions



Pair up and discuss



Write

Peer review: Pair up



- Read each other’s proposal and literature review.
- Most common experience: “Don’t understand (yet).”
- Your supervisor should only come after it generally makes sense.

How to determine the research question: Main type

Given what others have done, what remains to be done as next step(s)?
→ You derive the RQ from the gaps you have identified in literature.

Main Types of RQs	**Answers**
Exploratory	Answers give clearer understanding, better definitions of concepts, evidence that we can measure them validly
Base-rate	Answers describe the normal pattern of occurrence of the phenomena
Relationship	Answers describe if and how two phenomena are related
Solution-focused	Describes better ways to solve some problem or situation. Examples are: “What is an effective way to achieve X?”, “Which strategies help achieve X?”, “How can we refined S to achieve X in a better way?”

How to determine the research question: Sub-type

Sub-Types of RQs	Examples
Exploratory/Existence	“Does X exist?”, “Is Y something that software engineers really do?”
Exploratory/Descriptive	“What is X like?”, “What are its properties/attributes?”, “How can we categorize/measure X?”, “What are the components of X?”
Exploratory/Comparative	“How does X differ from Y?”
Base-rate/Frequency	“How often does X occur?”, “What is an average amount of X?”
Base-rate/Process	“How does X normally work?”, “What is the process by which X happens?”, “In what sequence does the events of X occur?”
Relationship/Existence	“Are X and Y related?”, “Do occurrences of X correlate with Y?” “What correlates with X?”
Relationship/Causality	“What causes X?”, “Does X cause Y?”, “Does X prevent Y?”,
Causality/Comparative	“Does X cause more Y than Z does?”, “Is X better at preventing Y than Z is?”
Causality/Context	“Does X cause more Y under one condition than others?”

How to determine the research method(s)



- How did other people tackle this based on your lit review?
- What makes sense to complement that to answer the gap?

- Also: Read the ABC of Software Engineering Research for overview*

*Klaas-Jan Stol and Brian Fitzgerald. 2018. The ABC of Software Engineering Research. ACM Trans. Softw. Eng. Methodol. 27, 3, Article 11 (September 2018), 51 pages. https://doi.org/10.1145/3241743